EXPERIMENT 3

AIM: Implement basic commands of linux like ls, cp, mv and others using kernel APIs

THEORY: The Linux kernel provides several interfaces to user-space applications that are used for different purposes and that have different properties by design. There are two types of application programming interface (API) in the Linux kernel that are not to be confused: the "kernel–user space" API and the "kernel internal" API.

The Linux API is the kernel–user space API, which allows programs in user space to access system resources and services of the Linux kernel.

Implement "cp" using system calls.

) First check if both source & the destination files are received from the command line argument and exit if argc counter is not equal to 3. There is also a check to handle "–help" option to print the usage of cpcmd.

3) Open the destination file with the respective flags & modes.

O_WRONLY -> Open the file in write only mode

O_TRUNC -> Truncates the contents of the existing file

O_CREAT -> Creates a new file if it doesn't exist

S_IXUSR are file permissions for the current user ('X' can be R for read & W for write)
S_IXGRP are file permissions for the groups ('X' can be R for read & W for write)

S_IXOTH are file permissions for the groups ('X' can be R for read & W for write)

4) Start data transfer from source file to destination file till it reaches EOF (nbread == 0). One of the most common Linux commands is the ls command. The "ls" is a Linus shell command which is used to print all the files and directories in the present directory in the form of a list.

READ FROM KEYBOARD:

```
#include<stdio.h>
```

```
#include<unistd.h>

int main()

{

int n;

char buff[20];

n=read(0,buff,10)

write(1,buff,n);

}
```

OUTPUT:

```
siesgst@siesgst-OptiPlex-3000:~$ gcc nread.c
siesgst@siesgst-OptiPlex-3000:~$ ./a.out
hello
hello
```

READ FROM FILE:

```
#include<stdio.h>

#include<sys/types.h>

#include<fcntl.h>

#include<unistd.h>

int main(){int n,f;

char buff[100];

f=open("name",O_RD

ONLY);
```

```
n=read(f,buff,50);

write(1,buff,n);}
```

OUTPUT:

```
siesgst@siesgst-OptiPlex-3000:~$ gcc nread.c
siesgst@siesgst-OptiPlex-3000:~$ gcc nread.c
siesgst@siesgst-OptiPlex-3000:~$ ./a.out
1 suraj K
2 deepak C
3 Aditya R
siesgst@siesgst-OptiPlex-3000:~$
```

WRITE THROUGH KEYBOARD:

```
#include<stdio.h>

#include<sys/types.h>

#include<fcntl.>

#include<unistd.h>

int main(){

int n,fd;

char buff[100];

fd =open("f1",O_WRONLY);

n=read(0,buff,50);

write(fd,buff,n);

}
```

OUTPUT:

```
siesgst@siesgst-OptiPlex-3000:~$ gcc nread.c
siesgst@siesgst-OptiPlex-3000:~$ ./a.out
Hello this is OS lecture
siesgst@siesgst-OptiPlex-3000:~$ cat f1
Hello this is OS lecture
siesgst@siesgst-OptiPlex-3000:~$
```

COPY FROM 1 FILE TO ANOTHER:

```
#include<stdio.h>

#include<sys/types.h

#include<fcntl.h>

#include<unistd.h>

int main(){

int n,fd1,fd2;

char buff[100];

fd1

=open("f1",O_RDONLY);

fd2=open("f2",O_WRONLY)

; n=read(fd1,buff,50);

write(fd2,buff,n);

}
```

OUTPUT:

```
siesgst@siesgst-OptiPlex-3000:~$ gcc nread.c
siesgst@siesgst-OptiPlex-3000:~$ ./a.out
siesgst@siesgst-OptiPlex-3000:~$ cat f1
Hello this is OS lecture
siesgst@siesgst-OptiPlex-3000:~$ cat f2
Hello this is OS lecture
siesgst@siesgst-OptiPlex-3000:~$
```

CONCLUSION: Thus, we have implemented basic commands of linux